FEC: A Feature Fusion Framework for SAR Target Recognition Based on Electromagnetic Scattering Features and Deep CNN Features

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***Abstract*— The active recognition of interesting targets has been a vital issue for synthetic aperture radar (SAR) systems. The SAR recognition methods are mainly grouped as follows: extracting image features from the target amplitude image or matching the testing samples with the template ones according to the scattering centers extracted from the target complex data. For amplitude image-based methods, convolutional neural networks (CNNs) achieve nearly the highest accuracy for images acquired under standard operating conditions (SOCs), while scattering center feature-based methods achieve steady performance for images acquired under extended operating conditions (EOCs). To achieve target recognition with good performance under both SOCs and EOCs, a feature fusion framework (FEC) based on scattering center features and deep CNN features is proposed for the first time. For the scattering center features, we first extract the attributed scattering centers (ASCs) from the input SAR complex data, then we construct a bag of visual words from these scattering centers, and finally, we transform the extracted parameter sets into feature vectors with the *k*-means. For the CNN, we propose a modified VGGNet, which can not only extract powerful features from amplitude images but also achieve state-of-the-art recognition accuracy. For the feature fusion, discrimination correlation analysis (DCA) is introduced to the FEC framework, which not only maximizes the correlation between the CNN and ASCs but also decorrelates the features belonging to different categories within each feature set. Experiments on Moving and Stationary Target Acquisition and Recognition (MSTAR) database demonstrate that the proposed FEC achieves superior effectiveness and robustness under both SOCs and EOCs.**

***Index Terms*— Convolutional neural network (CNN), electromagnetic scattering center, feature fusion, Synthetic aperture radar (SAR), target recognition.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active imaging system based on the coherent processing of multiple radar echoes, and it has the advantages of all-time operations, all-weather operations, and a long operating distance [1]. As a branch of machine vision, extending machine intelligence into SAR image processing has played crucial roles in military and civilian fields, e.g., terrain classification [2], [3], target detection [4], [5], change detection [6], and automatic target recognition (ATR) [7], [8]. Aiming at accurately detecting and recognizing targets from high-resolution SAR images, an ATR system can be broken down into three parts: detection, discrimination, and classification [7], [9]. The primary purpose of target classification is to automatically categorize each input image patch that is acquired by the target detection and discrimination. SAR target recognition has been refined as the electronics field has developed, such as with scattering theory, image processing, and machine vision. For an SAR target recognition system, the major challenge is the variability of the target signature caused by different SAR sensors and multiple operating conditions (OCs). The OCs can be categorized into the standard operating conditions (SOCs) and extended operating conditions (EOCs) [10]. SOCs mean that the testing and training images are acquired under either similar or even the same radar imaging conditions. However, for EOCs, the two types of images are obtained under different imaging conditions, e.g., different depression angles, different levels of noise interference, and complex ground backgrounds. Although many methods [11]–[14] can achieve very high accuracy in classifying images under SOCs, their accurate recognition of SAR images under EOCs is still a challenge.

An SAR target recognition method is usually composed of two parts: a feature-extraction module and a classifier module. The feature-extraction methods are mainly used to obtain the following features: geometric features, transform domain features, electromagnetic scattering features, and deep learning (DL) features.

1) For the geometric features, Clemente *et al.* [15] introduced Krawtchouk moments to describe SAR images' binary areas. Akbarizadeh [16] used the statistical-based kurtosis wavelet energy for feature extraction, which improves the segmentation accuracy without removing speckled noise. Park *et al.* [17] used the physical size of different targets. Raeisi *et al.* [18] combined the cuckoo search algorithm and nonnegative matrix factorization of Zernike moment features. Tirandaz and Akbarizadeh [19] designed an optimum kernel function using a parameter estimation algorithm based on the curvelet coefficient energy

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(KCE). It is noticeable that KCE can extract more statistical data from the textures of the SAR images than traditional feature extraction methods.

2) For the transform domain features, principal component analysis (PCA), linear discriminant analysis (LDA), and their variants [20], [21] are widely used to extract the features of SAR images. Meanwhile, signal analysis-based methods, e.g., the wavelet transform and Gabor filter [16], [22], also precisely recognize SAR targets.
3) The electromagnetic scattering features [7], [10], [23], a type of features that are extracted from the complex images of targets, represent a target's inherent physical properties, e.g., geometrical shapes, structures, and material information. Thanks to its property of invariance to azimuth angle variation and resolution differences, and robustness to signal-to-noise ratio (SNR) changes, electromagnetic scattering feature-based methods can recognize SAR targets with high accuracy, especially for images under EOCs [10], [23], [24].
4) DL, a widely used classification method in computer vision, is also used to extract the features of SAR images and recognize their targets. Composed of multiple convolution layers such as pooling layers and activation layers, convolutional neural networks (CNNs) nearly achieve the most accurate recognition in the Moving and Stationary Target Acquisition and Recognition (MSTAR) database under SOCs [11], [25]–[28]. However, since the features of the testing images differ from those of the training images under EOCs, in this situation, the CNN-based methods may be less likely applied to recognize SAR images [24]. Therefore, extracting the features of images under either SOCs or EOCs is still a vital issue in SAR target recognition.

Both extracting features and being an excellent classifier are necessary for supervised-based recognition methods. The most used classifiers in SAR target recognition are as follows.

1) *Nearest Neighbors (NNs) [29]:* Dungan and Potter [30] classified SAR images with transformation-variant attributed point patterns and the NNs. Ding *et al.* 31] realized decision fusion based on physically relevant features and the Hausdorff distance.
2) *Support Vector Machine (SVM):* Zhao and Principe [9] first introduced the SVM into SAR recognition, which was more accurate than other matching-based methods. Liu *et al.* [32] combined the wavelet transform and SVM with class-dependent structure-preserving projections.
3) *Spare Representation-Based Classification (SRC) [33]:* Song *et al.* [34] extracted multiple types of features from the MSTAR data set and trained an SRC with these features. Zhang *et al.* [35] used a joint sparse representation (SR) to classify multiview images. These methods try to establish different and discriminative classifiers based on one or more types of representative features. By and large, effectively combining feature-extraction methods and classifiers is very essential to recognizing SAR targets with high accuracy and robustness, especially under EOCs.

A. Background

In recent years, DL has been widely used in SAR image processing. For example, the deep belief network [6], [36] has been employed as the deep architecture in change detection, which improved the detection performance with a low computational volume. Kang *et al.* [37] proposed a feature fusion algorithm based on stacked autoencoder which performs high recognition accuracy. Among these DL methods, CNNs have achieved great success in SAR target recognition. Chen *et al.* [11] first proposed a fully convolutional network (A-ConvNet). It significantly decreases the quantity of weight parameters while achieving a recognition accuracy exceeding 99%. Morgan [38] effectively combined CNN and softmax classifier which can learn discriminative features from training data. Sharifzadeh *et al.* [39] suggested a hybrid CNN and multilayer perceptron algorithm (CNN–MLP) for image classification, which has better object classification performance than the state-of-the-art methods. Lin *et al.* [28] proposed a convolutional highway unit to train deeper networks with limited SAR data. Du *et al.* [40] trained a displacement- and rotation-insensitive CNN. Ding *et al.* [41] proposed an effective method that tries to augment training images using rotation, translation, and adding noise. Wagner [12] extracted the features from input images using a CNN and used an SVM as the classifier, which improved the recognition accuracy. Malmgren-Hansen *et al.* [14] used emulated SAR images to initialize a CNN and then trained this CNN with MSTAR images. The reason why CNNs are more accurate than other traditional methods is mainly because a well-trained CNN can extract more robust and representative features from input SAR amplitude images [11]. Due to the severe differences between images under SOCs and EOCs, the recognition accuracy of CNN-based methods severely decreases for images acquired under EOCs [10], [24]. One reason is because the input of the CNN is constructed by the amplitude images of SAR data without the phase information. Another reason may be that the convolution layers in CNNs based on amplitude images cannot digest the implicit relation between target characteristics and radar working parameters.

Unlike CNN-based methods that extract contour features from amplitude images, electromagnetic scattering feature-based methods describe the scattering centers of SAR complex images from the perspective of SAR imaging [7], [24]. Scattering center features reflect the local electromagnetic phenomenon of radar targets. Extracting the scattering center can be regarded as optimizing the parameters of different scattering center models, such as the geometrical theory of diffraction (GTD) and the attributed scattering center (ASC) [42]. The extracted scattering centers are usually determined by the space position, complex intensity, and other physical parameters. There are already SAR target recognition methods based on scattering centers. Chiang *et al.* [43] proposed a model-based Bayesian matching method for SAR recognition. Dungan and Potter [30] used the least trimmed square Hausdorff distance (LTS-HD) to match the scattering center sets. Ding *et al.* [10] proposed a Hungarian-based matching method that is stable and achieves high recognition accuracy for images under EOCs. These position matching-based methods

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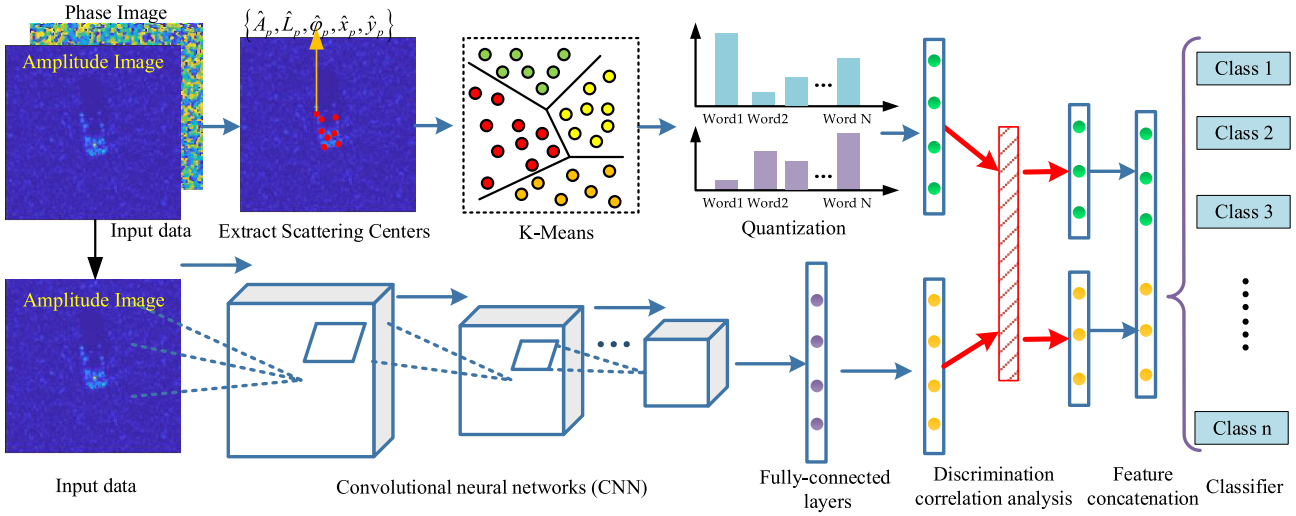


Fig. 1. Structure of the proposed FEC framework, which works by fusing the electromagnetic scattering features and deep CNN features.

work by matching the scattering centers of training data and testing data, but their recognition performance is easily affected by variations in the training samples. Meanwhile, the extracted electromagnetic scattering features are composed of multiple parameter sets, while other types of features are usually represented as feature vectors. This difference makes it difficult to fuse the electromagnetic scattering features and other kinds of features, and thus this restricts the performance of SAR target recognition.

In a nutshell, CNN feature-based methods achieve very high accuracy for images acquired under SOCs, while scattering center feature-based methods perform strongly and are stable for images acquired under EOCs. If these two features can be effectively fused, the recognition performance is believed to improve for SAR images acquired under SOC and EOCs.

B. Related Work

There are a few articles [17], [24], [25], [37], [44] that try to fuse multiple features to improve the recognition accuracy for SAR targets. Most of these articles pay more attention to extracting image features, e.g., PCA features and wavelet features, from the amplitude images of SAR targets. The effectiveness of electromagnetic scattering center features was ignored in these methods. To make use of complex images with interesting targets, Ding *et al.* [24] proposed a novel SAR target recognition framework that effectively fuses global and local features. In this framework, the nonparametric random projection (RP) features are taken as the global features and the ASC features are taken as the local features. The decision engine in this framework first is used to determine the test image as an "in-set" target or an outlier based on the confidence values of SRCs and then to further make a decision for the potential "in-set" target based on the ASC. Therefore, this decision-level fusion strategy is a type of "late fusion." However, the feature level fusion is believed to be more effective than this late fusion owing to the fact that a feature set contains more abundant information about the input image data than the matching score or the output decision of a classifier [45]. Although this framework performs highly accurate recognition for SAR images under SOCs and EOCs, the fusion strategy can still be enhanced.

C. Contributions

A novel SAR target recognition framework (named FEC) is proposed in this article. It works by fusing electromagnetic scattering features and deep CNN features. As shown in Fig. 1, the FEC method is composed of three modules: a feature extraction module based on scattering centers, a feature extraction module based on the deep CNN, and a feature fusion module. Compared with other SAR target recognition methods, FEC makes use of the stability of the scattering center features under EOCs and the powerful feature representation ability of CNN-based methods under SOCs. It is noticeable that this is the first time that the electromagnetic scattering center features and CNN features have been fused for SAR target recognition at the feature level. The main contributions of FEC are shown as follows.

1) The bag of visual words (BOVW) is introduced into the ASC model. This method can not only capture the stabile and sparse scattering center features from the input complex data but also convert the extracted features into a set of feature vectors with a fixed dimension. Compared with other recognition methods based on scattering center features, the proposed method is free from matching the complicated scattering centers in either the training samples or the testing samples. Furthermore, the extracted feature vectors can be readily used in the feature fusion module and classifier module at once.
2) A lightweight modified VGGNet (MVGGNet) is proposed for extracting deep CNN features from input amplitude images. Since the size and quantity of SAR images are not as large as those of optical images, the learning convergence and recognition performance of typical CNNs, e.g., VGGNet [46] and Resnet 47], are dissatisfactory for SAR image classification. To tackle these issues and extract a CNN feature that is consistent

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with the ASC feature, the number of fully connected layers and neurons of the original VGG-16 net are reduced according to the dimension of the ASC feature. The probability to correctly classify the proposed MVGGNet is 99.27%, which has achieved state-of-the-art performance.

3) A novel feature fusion method based on discrimination correlation analysis (DCA) is introduced to combine the relevant information extracted from ASC features and CNN features. Compared with other methods that fuse the input feature vectors without preprocessing any feature, this method makes uses of the between-class correlations and within-class correlations of ASC features and CNN features. Furthermore, compared to nonfused features, the fused features can be easily combined with typical classifiers due to their smaller dimension.

The rest of this article is organized as follows. The proposed FEC method is described in Section II. Section II first introduces the scattering center feature extraction method based on the BOVW, then presents the MVGGNet, and further proposes the feature fusion method based on discriminant correlation analysis. The experimental results on the MSTAR data are analyzed in Section III. Finally, the conclusions are drawn in Section IV.

II. Methodology

We will introduce the structure of FEC in three parts.

A. Scattering Center Feature Extraction Based on the BOVW

Most SAR target recognition methods based on scattering center features work by matching the extracted scattering centers of testing images and template images [7], [10], [43]. Thus, the scattering center features extracted by these methods cannot be directly fused with other types of features at the feature level. We need to convert the set of scattering center features into a feature vector to fuse them. Therefore, the BOVW of scattering center features is proposed in this section.

1) Simplified ASC Model: The radar echoes of an SAR system can be modeled as a superposition of multiple scattering centers [42]

$$\mathbf{S}(f,\varphi)=\sum S_p(f,\varphi)+e(f,\varphi) \tag{1}$$

where $S_p(f,\varphi)$ is the echo data of the scattering center p, $\mathbf{S}(f,\varphi)$ is the entire echo of all scattering centers, and $e(f,\varphi)$ represents the noise and clutter in the imaging process. f and φ are the frequency and azimuth angle, respectively. The scattering center can be modeled as a GTD parametric model or an ASC model, among other potential models. Due to the powerful physical representation capability of the ASC model, the scattering center p in (1) obeys this ASC model

$$\begin{aligned}S_p(f,\varphi)=&A_p\left(j\frac{f}{f_c}\right)^{\alpha_p}\sin c\left[2\pi\frac{L_p}{c}f\sin\left(\varphi-\bar{\varphi}_{0p}\right)\right]\\&\times\exp\left(-2\pi\gamma_p f\sin\varphi\right)\\&\times\exp\left[-j4\pi\frac{f}{c}\left(x_p\cos\varphi+y_p\sin\varphi\right)\right]\end{aligned} \tag{2}$$

where $\sin c(\cdot)=\sin(\cdot)/(\cdot)$, f_c is the radar center frequency, and c is the speed of light. $\Theta_p=[A_p,\alpha_p,x_p,y_p,L_p,\bar{\varphi}_{0p},\gamma_p]$ is the parameter set of the scattering center p. A_p is the complex amplitude, and α_p is the frequency dependence that falls into $[-1,-0.5,0,0.5,1]$. x_p and y_p are, respectively, the length and orientation angle of the distributed scattering center p. γ_p is the azimuth dependence. According to [48], when the ratio of the radar system bandwidth to the center frequency is small, the frequency-dependent factor α_p could be ignored. Meanwhile, γ_p is usually small for an SAR system. Therefore, the original ASC model can be simplified as follows:

$$S_p(f,\varphi)=A_p\phi_p \tag{3}$$

where

$$\begin{aligned}\phi_p(f,\varphi)=&\sin c\left[2\pi\frac{L_p}{c}f\sin\left(\varphi-\bar{\varphi}_{0p}\right)\right]\\&\times\exp\left[-j4\pi\frac{f}{c}\left(x_p\cos\varphi+y_p\sin\varphi\right)\right].\end{aligned} \tag{4}$$

To estimate the parameter set Θ_p of the scattering center p, $\phi_p(f,\varphi)$, $S_p(f,\varphi)$, and $\mathbf{S}(f,\varphi)$ are transformed into vectors $\vec{\phi}_p$, $\vec{S}_p$, and $\vec{S}$, respectively [48]. It is noticeable that $\vec{\phi}_p^{\,H}\vec{\phi}_p=1$ here. This parameter estimation can be represented as follows:

$$\hat{\Theta}_p=\left\{\hat{A}_p,\hat{L}_p,\hat{\varphi}_p,\hat{x}_p,\hat{y}_p\right\}_p=\arg\min_{\Theta_p}\left\|\vec{S}-\sum_p A_p\vec{\phi}_p\right\|^2. \tag{5}$$

The detailed optimization can be found in [48]. After substituting the estimated parameter set $\hat{\Theta}_p$ into (3), the signal of the reconstructed scattering center p is as follows:

$$\begin{aligned}\hat{S}_p(f,\varphi)=&\hat{A}_p\cdot\sin c\left[2\pi\frac{\hat{L}_p}{C}f\sin\left(\varphi-\hat{\bar{\varphi}}_{0p}\right)\right]\\&\times\exp\left[-j4\pi\frac{f}{C}\left(\hat{x}_p\cos\varphi+\hat{y}_p\sin\varphi\right)\right].\end{aligned} \tag{6}$$

The ASCs of measured BMP2 data from the MSTAR data set are shown in Fig. 2. It can be seen that the target characteristics can be precisely parameterized through the simplified ASC model. The noise and clutter are also removed in the reconstruction results. Meanwhile, ASC can perform stable feature extraction for SAR images with different SNRs [24].

2) Constructing Visual Vocabulary of ASCs With k-Means: Through the simplified ASC model and corresponding estimation methods, SAR target complex data are represented as a few scattering centers with parameters $\{\hat{A}_p,\hat{L}_p,\hat{\varphi}_p,\hat{x}_p,\hat{y}_p\}$. Although this type of feature extraction method is similar to the scale-invariant feature transform (SIFT) [49] method, the SIFT method describes images' contour features with gradient amplitudes in multiple directions, while the ASC model describes SAR complex data with a few parameter sets. The BOVW based on the SIFT method has achieved very high accuracy at classifying optical images [50], [51] because the unsupervised k-means strategy in this method could rapidly find the optimal clustering center in each gradient direction. Meanwhile, this strategy is also favorable for the design of subsequent classifiers.

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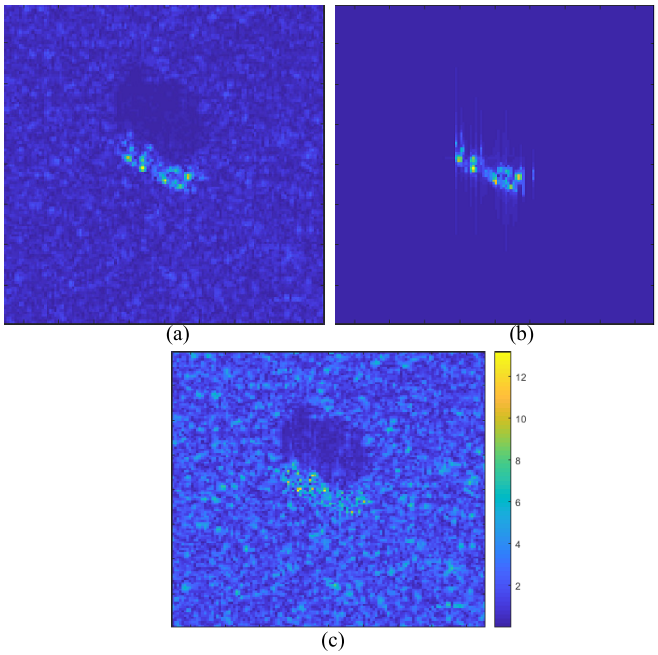


Fig. 2. ASC parameter extraction results and reconstruction results. (a) Amplitude image of the input complex data. (b) Reconstruction result of the simplified ASC model. (c) Residue image between (a) and (b).

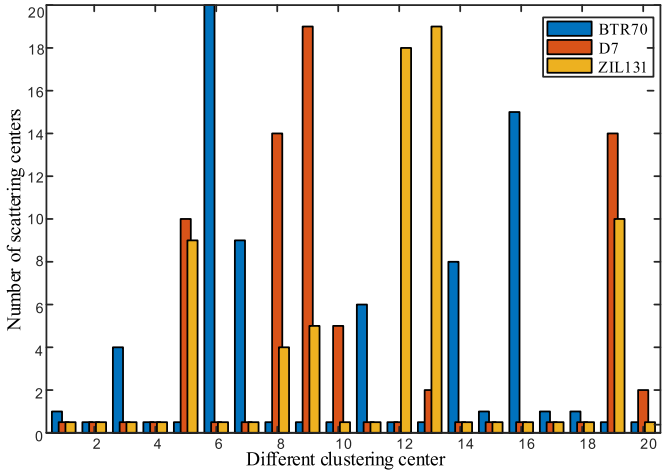


Fig. 3. Average number of scattering centers allocated to each clustering center. To intuitively understand the frequency distribution of different kinds of targets, three types of targets are chosen from the MSTAR data set. The number of scattering centers extracted from each input data is set as 70, and the number of clustering centers is set as 20.

Inspired by the BOVW, the k-means is introduced in this article to search the clustering center of each parameter in the parameter set $\{\hat{A}_p, \hat{L}_p, \hat{\varphi}_p, \hat{x}_p, \hat{y}_p\}$. For a pair of real vectors $X = (x_1, x_2, \ldots, x_n)$ and $Y = (y_1, y_2, \ldots, y_n)$, the distance of the original k-means in Euclidean space is

$$d = \sqrt[q]{|x_1 - y_1|^q + |x_2 - y_2|^q + \cdots + |x_n - y_n|^q} \tag{7}$$

where q means the distance order. For an estimated parameter set of ASC, all parameters in $\hat{\Theta}$ can be measured in (7) except for the complex amplitude $\hat{A}$. To measure $\hat{A}$, it is decomposed into an amplitude part $|\hat{A}|$ and a phase part $\omega^{\hat{A}}$, where $\omega^{\hat{A}}$ falls into $[0, 2\pi]$. Thus, for the scattering center p, $\hat{\Theta}_p$ can be written as $\{|\hat{A}_p|, \omega_p^{\hat{A}}, \hat{L}_p, \hat{\varphi}_p, \hat{x}_p, \hat{y}_p\}$. To determine the number of clustering centers, the k-means is run several times in this article with different numbers of desired representative vectors and different sets of initial cluster centers [51].

3) Scattering Center Feature Vectorization: With the above k-means processing, we get a visual vocabulary $\{\theta_1^c, \theta_2^c, \ldots, \theta_N^c\}$, which is composed of multiple clustering centers. To transform the extracted ASC features into a fixed-dimension vector, the quantization of the scattering center p is done as follows:

$$k = \arg\min_i \left\|\theta_p - \theta_i^c\right\|^2 \tag{8}$$

where $i \in \{1, 2, \ldots, N\}$. For the measured SAR complex data, assuming that the number of extracted scattering centers is M, these M scattering centers are assigned to N different clustering centers. Counting the number of scattering centers allocated to each clustering center and regularizing by M, the input SAR data are converted into an N-dimension vector. For this feature vector, targets of the same category should be subject to the same or a similar frequency distribution. As shown in Fig. 3, the quantity distribution of scattering centers of BTR70, D7, and ZIL131 allocated to each clustering center performs completely different.

Most SAR target recognition methods based on scattering center features work by matching the scattering centers in pixel space [7], [10], [43]. Unlike these methods, this article tries to transform SAR complex data into a vector through the BOVW method. This type of extracted feature can not only be used for the standard classifiers, e.g., the random forest (RF) and SVM, but can also be fused with other features, e.g., CNN features in this article.

B. High-Dimensional Feature Extraction Based on the MVGGNet

CNN, a typical deep neural network, has been widely used in optical image classification. There are lots of CNNs, e.g., Alexnet [52], VGGNet [46], and Resnet [47], which can perform highly accurate classification for published optical data sets. A direct way to represent features is to train an SAR target recognition model with the above typical networks. However, there still remain two difficulties when this type of method is applied. One is that the smaller size of SAR target images and more pooling layers in these networks will cause too much resolution loss, thus affecting the recognition accuracy. The other is that the lower number of SAR images will affect the training convergence and classification accuracy. Since VGGNet can successfully extract the features in various tasks [3], [53], [54], the VGG-16 net is taken as the basic feature extraction network, and we have modified the fully connected layers of this net.

The structure of the MVGGNet is shown in Table I. "Type" means different network layers, e.g., convolution layer ("Conv"), pooling layer ("Pool"), fully connected layer ("FC"), and dropout layer ("Drop"). The "size," "number," and "stride" in Table I are the kernel settings in different layers. "Output" is the size of the output

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TABLE I
NETWORK STRUCTURE OF THE MVGGNET

Layer	Type	Size	Number	Stride	Output	Parameter
0	Input	—	—	—	128×128	—
1	Conv	3×3	64	1×1	128×128	1,728
2	Conv	3×3	64	1×1	128×128	36,864
3	Pool	2×2	—	2×2	64×64	—
4	Conv	3×3	128	1×1	64×64	73,728
5	Conv	3×3	128	1×1	64×64	147,456
6	Pool	2×2	—	2×2	32×32	—
7	Conv	3×3	256	1×1	32×32	294,912
8	Conv	3×3	256	1×1	32×32	589,824
9	Conv	3×3	256	1×1	32×32	589,824
10	Pool	2×2	—	2×2	16×16	—
11	Conv	3×3	512	1×1	16×16	1,179,648
12	Conv	3×3	512	1×1	16×16	2,359,296
13	Conv	3×3	512	1×1	16×16	2,359,296
14	Pool	2×2	—	2×2	8×8	—
15	Conv	3×3	512	1×1	8×8	2,359,296
16	Conv	3×3	512	1×1	8×8	2,359,296
17	Conv	3×3	512	1×1	8×8	2,359,296
18	Pool	2×2	—	2×2	4×4	—
19	FC		8192×256		256	2,097,152
20	Drop		Rate=0.5		256	—
21	FC		256×10		10	2,560

feature maps of the current layer. "Parameter" means the number of weighting parameters in the convolutional kernels while ignoring the bias parameter. The layers from layer 1 to layer 18 of this net keep the same structure with the original VGG-16 net. When using this network, the input SAR image with a size of 128 × 128 pixels is converted into a group of 512 feature maps that are sized 4 × 4 pixels. Therefore, the total stride of this network is 32 × 32 pixels.

The main difference between the original VGG-16 and the MVGGNet is the fully connected layers. The original VGG-16 net is devised for the classification of optical image data sets, e.g., ImageNet. A large number of neurons, e.g., 4096, in the fully connected layers of the original VGGNet could map the original input images to a high-dimensional feature space, where different types of objects can be effectively discriminated. It is noticeable that the number of classes of SAR targets in the published data set is much smaller than that of an optical data set [55], which means that the features of these SAR targets can also be well separated in a low-dimension feature space. Therefore, if the original VGG16 net is directly used for the recognition of SAR images, a large number of neurons in fully connected layers not only lead to too great of a computation load but also raise the risk of overfitting in the training process. As shown in Table I, the three cascaded fully connected layers in the original VGG16 net are reduced to two layers. The two cascaded fully connected layers with fewer neurons are named by the MVGGNet. This modification cuts down the number of weights of the fully connected layers from 50 372 608 to 2 099 712, thus maximally utilizing the small samples.

To train this MVGGNet, transfer learning is used in this article [14], [56]. More specifically, we first initialize the weighting parameters of layer 1 to layer 18 with the model

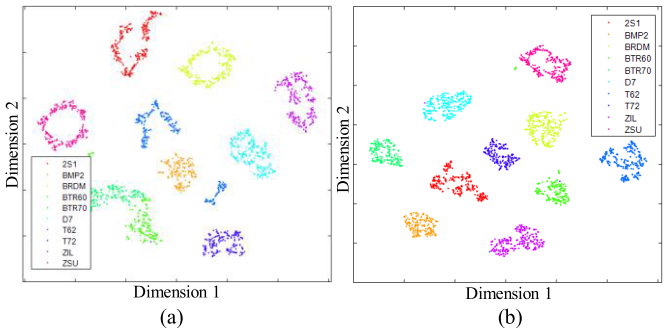


Fig. 4. Visualized features extracted by the original VGGNet and MVGGNet. Each scatter in these figures represents a testing sample in the MSTAR data set. (a) 4096-D feature of the second fully connected layer in the original VGGNet. (b) 256-D feature of the first fully connected layer in the MVGGNet.

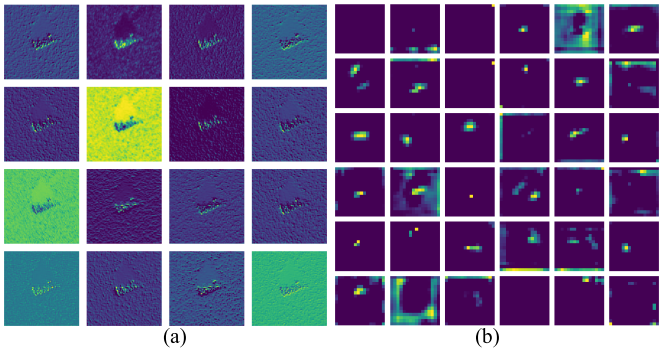


Fig. 5. Output feature maps of different layers of the MVGGNet when given a measured T72 target. (a) Output feature maps of convolution layer 5. (b) Output feature maps of convolution layer 13.

trained on the ImageNet data set, and then we train this MVGGNet with the training samples from the MSTAR data set. The average classification accuracy under the SOCs of this modified model reaches 99.27%, which already achieves state-of-the-art performance. The setup for the training can be found in the Section III. To visualize the high-dimensional features of the original VGGNet and MVGGNet, the t-SNE method [57] is used to reduce the dimensionality of these features to 2-D and embed the nearby points in the high-dimensional space to nearby low-dimensional points. The visualized features in Fig. 4 show that although the number of fully connected neurons in the MVGGNet is less than that of the original VGGNet, different types of targets can be easily separated by the MVGGNet. It is noticeable that the extracted 256-D feature vectors are used as the second type of feature in subsequent feature fusion. The extracted feature maps of different layers of the MVGGNet are shown in Fig. 5. One can see that the well-trained MVGGNet can extract powerful features from input SAR images, e.g., contour features in Fig. 5(a) and keypoint features in Fig. 5(b).

C. Target Recognition Based on Fused Features

In SAR target recognition, we combine the relevant information from the extracted ASC features and the CNN features in order to fuse the features. This method is expected to be more discriminative than when using input feature vectors [45], [58].

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Some typical feature fusion methods, e.g., serial feature fusion and parallel feature fusion, try to fuse the input feature vectors without preprocessing the features. To make use of the between-class correlations and within-class correlations, DCA [45] is introduced to this target recognition framework.

1) Feature Transformation Based on Discriminant Correlation Analysis: Assume that c is the number of categories belonging to the image data set D. For a feature matrix $X \subseteq \mathbb{R}^{p\times n}$ of D, p is the dimension of the features, and n is the number of samples in D. The n columns of X are composed of c separate classes, where n_i columns belong to the ith class ($n = \sum_{i=1}^{c} n_i$). $x_{i,j} \in X$ denotes the feature vector extracted from the jth image data in the ith category. $\bar{x}_i$ and $\bar{x}$ denote the means of the $x_{i,j}$ vectors in the ith class and the whole feature set, respectively, where $\bar{x} = (1/n)\sum_{i=1}^{c} n_i \bar{x}_i$ and $\bar{x}_i = (1/n_i)\sum_{j=1}^{n_1} x_{i,j}$. The between-class scatter matrix is defined as follows:

$$S_{bx(p\times p)} = \Phi_{bx}\Phi_{bx}^{\mathrm{T}} \tag{9}$$

where $\Phi_{bx(p\times c)} = [(n_1)^{1/2}(\bar{x}_1 - \bar{x}), (n_2)^{1/2}(\bar{x}_2 - \bar{x}), \ldots, (n_c)^{1/2}(\bar{x}_c - \bar{x})]$.

If the targets of different classes can be well separated, $\Phi_{bx}^{\mathrm{T}}\Phi_{bx}$ should be a diagonal matrix [45] as follows:

$$\varphi^{\mathrm{T}}\left(\Phi_{bx}\Phi_{bx}^{\mathrm{T}}\right)\varphi = \Lambda_{(r\times r)} \tag{10}$$

where Λ is the diagonal matrix of real and nonnegative eigenvalues sorted in decreasing order, and φ consists of the first r eigenvectors that correspond to the r largest nonzero eigenvalues.

The transformation of S_{bx} is $W_{bx} = \Phi_{bx}\varphi\Lambda^{-1/2}$, which can reduce the dimensionality of feature matrix X from p to r

$$X'_{(r\times n)} = W_{bx(r\times p)}^{\mathrm{T}} X_{(p\times n)}. \tag{11}$$

Therefore, W_{bx} projects X to X' in a special space, where the between-class scatter matrix is I and the targets of different classes are separated. Similar to X, the transformation of the feature matrix $Y \subseteq \mathbb{R}^{q\times n}$ is as follows:

$$Y'_{(r\times n)} = W_{by(r\times q)}^{\mathrm{T}} Y_{(q\times n)} \tag{12}$$

where $r \le \min(c-1, \mathrm{rank}(X), \mathrm{rank}(Y))$.

To make the features in one set have nonzero correlation only with their corresponding features in the other set, the between-set covariance matrix of the transformed feature sets $S'_{xy} = X'Y'^{\mathrm{T}}$ should be diagonalized using singular value decomposition (SVD)

$$S'_{xy(r\times r)} = U\sum V^{\mathrm{T}}. \tag{13}$$

Let $W_{cx} = U\sum^{-1/2}$ and $W_{cy} = V\sum^{-1/2}$. Then, the feature sets are transformed as follows:

$$X^* = W_{cx}^{\mathrm{T}}X' = W_{cx}^{\mathrm{T}}W_{bx}^{\mathrm{T}}X = W_x X \tag{14}$$

$$Y^* = W_{cy}^{\mathrm{T}}Y' = W_{cy}^{\mathrm{T}}W_{by}^{\mathrm{T}}X = W_y X. \tag{15}$$

With the above transformations, DCA does not only maximize the correlation of the corresponding features across the two feature sets, e.g., CNN features and ASC features, but it also decorrelates features that belong to different categories within each feature set.

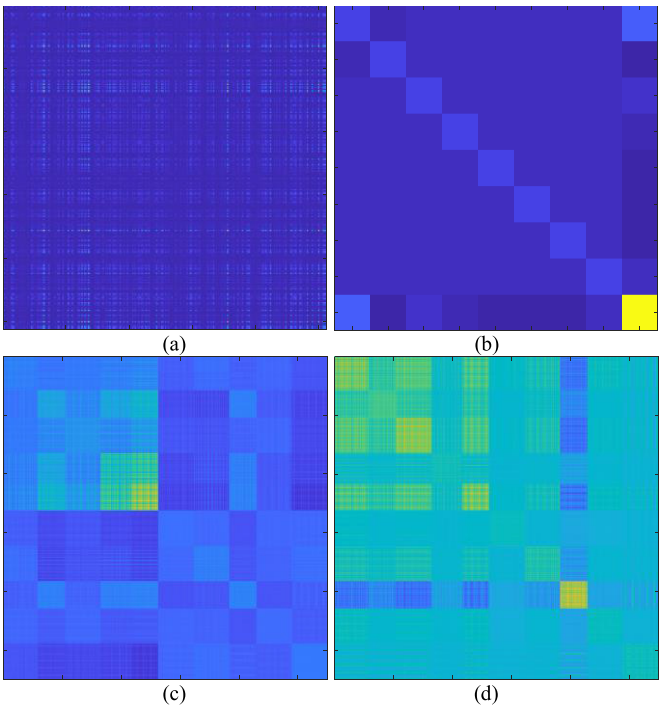


Fig. 6. Visualization of covariance matrices (dark color represents the minimum value and the elements with higher values are illustrated brighter). (a) Covariance between unconverted features XX^{T}. (b) Covariance between converted features X^*X^{*T}. (c) Covariance between unconverted samples $X^{\mathrm{T}}X$. (d) Covariance between converted samples $X^{*T}X$.

The results in Fig. 6(a) and (b) show that the covariance between different features within a feature set is reduced by the DCA transformation. The results in Fig. 6(c) and (d) show that the samples with DCA processing have a higher correlation only with the ones in the same class.

2) Target Recognition Based on Fused Features: After transforming the extracted ASC features and CNN features, the two types of features should be fused as one feature set. There are two typical fusion methods: concatenation and summation [58]

$$Z_1 = \begin{pmatrix} X^* \\ Y^* \end{pmatrix} = \begin{pmatrix} W_x^{\mathrm{T}} & X \\ W_y^{\mathrm{T}} & Y \end{pmatrix} = \begin{pmatrix} W_x & 0 \\ 0 & W_y \end{pmatrix}\begin{pmatrix} X \\ Y \end{pmatrix} \tag{16}$$

$$Z_2 = X^* + Y^* = W_x^{\mathrm{T}}X + W_y^{\mathrm{T}}X = \begin{pmatrix} W_x \\ W_y \end{pmatrix}^{\mathrm{T}}\begin{pmatrix} X \\ Y \end{pmatrix} \tag{17}$$

where Z_1 and Z_2 are the fused features. In this article, X and Y represent the CNN features and ASC features extracted from SAR data, respectively.

Since the summation strategy fused the two types of feature vectors into one unified feature vector, the feature dimension will be reduced by half, further causing the loss of feature information. Meanwhile, there are great differences between these two types of features: ASC features represent the physical characteristics of input SAR complex data, while the CNN features represent the contour and semantic features of the input amplitude image. To make use of these different feature sets, the CNN and ASC features are fused using the concatenation strategy in the FEC method.

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TABLE II
TRAINING AND TESTING IMAGES FOR THE SOC EXPERIMENTAL SETUP

Class	Serial Number	Training		Testing	
		Depression	Number	Depression	Number
BMP2	9563	17°	233	15°	195
BTR70	c71	17°	233	15°	196
T72	132	17°	232	15°	196
T62	A51	17°	299	15°	273
BRDM2	E-71	17°	298	15°	274
BTR60	7532	17°	256	15°	195
ZSU23/4	d08	17°	299	15°	274
D7	13015	17°	299	15°	274
ZIL131	E12	17°	299	15°	274
2S1	B01	17°	299	15°	274

When getting the fused feature vectors of the training samples and testing samples, the classifiers can be trained with the training feature vectors. The inference results can also be attained by inputting the testing feature vectors into a well-trained classifier. There are lots of classifiers for SAR target recognition, e.g., the SVM [59], K-nearest neighbors (KNN) [29], SRC [34], and RF [60]. Due to their varying sensitivity to input features, these classifiers obtain different recognition accuracies. Since the RF classifier can perform well for low-dimension feature vectors and a larger number of samples, it is used as the classifier of the FEC method. The experimental results in Section III prove the effectiveness of the combination of fused features and RF classifiers.

III. EXPERIMENTS AND ANALYSIS

A. Data Description

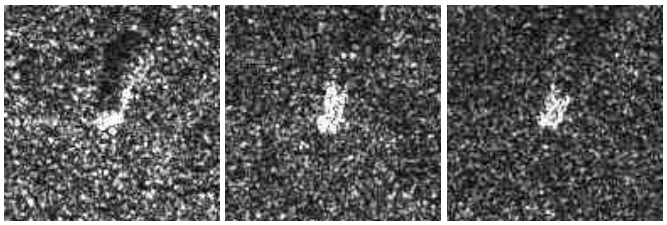
To evaluate the proposed FEC method, the MSTAR data set collected by the Sandia National Laboratory SAR sensor platform is used as a benchmark in this article. There are ten categories of ground targets in this data set: BMP2, BTR70, T72, T62, BRDM2, BTR60, ZSU23/4, D7, ZIL131, and 2S1. These images were acquired using an X-band SAR system with a resolution of 0.3 m $\times$ 0.3 m, a size of nearly 128 $\times$ 128 pixels, and aspect angles from 0° to 360°. Each datum in MSTAR is a complex image that can be decoupled into an amplitude fraction and a phase fraction. Unlike most articles [11], [12], [25], [26] that merely use amplitude images to classify targets, here both the amplitude images and phase images of the MSTAR data are used as the input of the proposed FEC method. Varying OCs will lead to different image distributions. The data descriptions of MSTAR under SOCs and EOCs are as follows.

1) SOC Data Description: SOCs mean that the testing images and training images are acquired under similar imaging conditions. The two types of images slightly differ in their depression angles. The depression angle of the training images is 17°, while it is 15° for the testing ones, as shown in Table II. The number of training images is 2747, and the number of testing images is 2425.

2) EOCs' Data Description: EOCs mean that the imaging conditions of testing images differ from those of the training ones, e.g., different depression angles, various levels of noise interference, and different configuration variants. The EOCs in MSTAR are as follows.

TABLE III
TRAINING AND TESTING IMAGES FOR THE EOC-1 EXPERIMENTAL SETUP

Class	Serial Number	Depression			
		Training	Testing		
		17°	15°	30°	45°
2S1	B01	299	274	288	303
BRDM2	E-71	298	274	287	303
ZSU23/4	d08	299	274	288	303



(a) (b) (c)

Fig. 7. BRDM2 images at different depressions. (a) 17°. (b) 30°. (c) 45°.

TABLE IV
TRAINING AND TESTING IMAGES FOR THE EOC-2 EXPERIMENTAL SETUP

Class	Serial Number	Depression			
		Training	Testing		
		17°	15°	17°	Total
BMP2	9563	233	-	-	-
	9566	-	196	232	428
	c21	-	196	233	429
BRDM2	E-71	298	-	-	-
BTR70	c71	233	-	-	-
T72	132	232	-	-	-
	812	-	195	231	426
	A04	-	274	299	573
	A05	-	274	299	573
	A07	-	274	299	573
	A10	-	271	296	567

a) EOC-1 (large depression variant): Since the SAR images are sensitive to the imaging depression, large depression differences will cause a great difference in the corresponding SAR images. The large depression variance is shown in Table III. The 2S1, BRDM2, and ZSU23/4 are chosen as the training data and testing data. The BRDM2 images at different depressions are shown in Fig. 7. One can see that there are some differences in the contour features and scattering center feature between Fig. 7(a) and (c). Therefore, it is crucial to test the recognition algorithms using large depression variances.

b) EOC-2 (version variant): Both BMP2 and T72 in the MSTAR data have several versions, as shown in Table IV. Different versions of the same class have very similar global structures, although there is some difference between their local structures, e.g., fuel tank and battery. Three versions of the T72 target are shown in Fig. 8(a)–(c). Evaluating the recognition accuracy of different versions of the target is meaningful for understanding the robustness of different methods.

c) EOC-3 (noise corruption): The noises caused by the radar system and the background will affect the measured SAR images and further affect the classification accuracy. The SNR of MSTAR images, which is usually at least 30 dB, seems unrealistic for real applications [24]. Therefore, we manually add the complex additive white Gaussian noise

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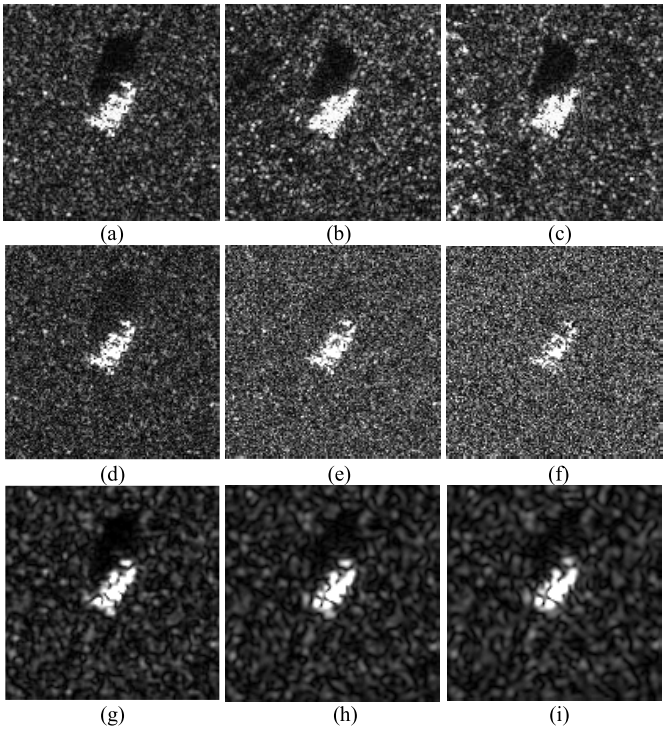


Fig. 8. T72 amplitude images with different EOCs. (a)–(c) Different versions of T72. (d)–(f) Different SNRs. (g)–(i) Different resolutions. (a) Version A04. (b) Version A05. (c) Version A07. (d) SNR 5 dB. (e) SNR 0 dB. (f) SNR −5 dB. (g) 0.4 m × 0.4 m. (h) 0.5 m × 0.5 m. (i) 0.6 m × 0.6 m.

to the original SAR data. The noisy corrupted T72 images at different SNRs are shown in Fig. 8(d)–(f).

d) EOC-4 (resolution variance): The resolution of SAR images will be different if we adjust the radar working parameters, e.g., working bandwidth and airplane flying speed. The image features will vary with the resolution change, further affecting the recognition performance. To evaluate the recognition results, images with different resolutions are simulated by segmenting different proportions of the frequency data corresponding to complex images [10], [24]. These images are interpolated to have the same pixel spacing as the original ones. The simulated amplitude images of T72 at different resolutions are shown in Fig. 8(g)–(i). One can see that the local features, e.g., contour features of the target component, become blurred as the resolution decreases.

B. Experimental Setup

The proposed FEC method consists of three parts: a feature extraction module based on scattering centers, a feature extraction module based on a deep CNN, and a feature fusion module. The parameter determination and training strategy of these modules are, respectively, given as follows.

There are two hyperparameters in the feature extraction of the scattering center: the number of scattering centers M extracted from each input SAR datum and the number of clustering centers N in the k-means. M is set as 70 in this article since it can well represent the entire target and effectively ignore the background clutter in the input complex data [61]. To determine N, an RF classifier is used to classify the ASC features. To be specific, we first choose different

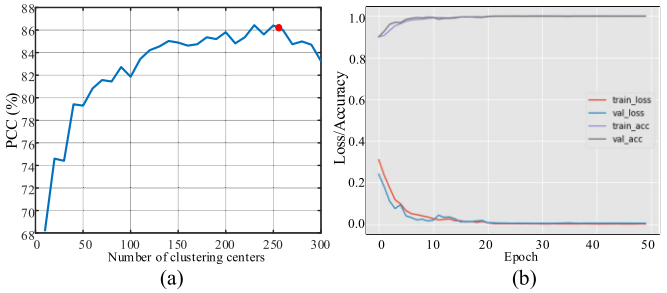


Fig. 9. Experimental results of ASC features and deep CNNs. (a) PCC with the number of clustering centers. (b) Training loss/accuracy and testing loss/accuracy of the MVGGNet.

numbers of clustering centers, then use the features extracted from the training samples to train the RF classifier, and finally determine the optimal number of clustering centers according to the probability of correct classification (PCC) of the testing samples. The number of trees is 200 and the depth of the trees is 10 for the RF classifier. The PCCs with different N s are shown in Fig. 9(a). It is noticeable that the PCC can also be improved by increasing the depth and number of decision trees in the RF classifier. To make use of the ASC features and CNN features and to keep their dimensions the same, N is set as 256 in this article.

For the feature extraction based on the deep CNN, the MVGGNet should be retrained because we cannot directly adopt most public VGG models that are trained with optical images. We add a softmax layer onto the last fully connected layers of the MVGGNet as the classifier. For the training process, the cross-entropy loss is used as the loss function, and the stochastic gradient descent (SGD) with an initial learning rate of 10^{-3} is used for the loss optimization. The batch size is set as 128, and the number of epochs is set as 50. The initial weight parameters of the MVGGNet are set by the VGG model that is well trained with optical images, which is a special form transfer learning [14], [56]. The training loss/accuracy and testing loss/accuracy based on the training iterations are shown in Fig. 9(b). The PCC of the well-trained MVGGNet is 99.27%, which achieves state-of-the-art performance. When getting the well-trained MVGGNet model, the activation value of the final fully connected layer corresponding to each input image is taken as the CNN feature vector of this image.

The extracted 256-D ASC feature and 256-D CNN feature are transformed into more powerful feature vectors by the DCA method in Section II-C. The concatenation strategy is used to fuse these two types of features. Based on the above steps, SAR data are converted into a feature vector. The feature vectors of the training samples are used to train a classifier and the feature vectors of the testing samples are used to test the classifier. The RF classifier is used to classify the fused features. The number and depth of the decision trees are 300 and 12, respectively. Then, the category of the testing samples can be determined by the well-trained FEC model.

C. Results Under SOCs

The training data and testing data in Table II are used as the data source of SOCs. Since a different size of the data patch could affect the feature extraction of the ASC and deep

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TABLE V
CONFUSION MATRIX OF THE PROPOSED FEC MODEL UNDER SOCS

Class	2S1	BMP2	BRDM2	BTR60	BTR70	D7	T62	T72	ZIL131	ZSU23/4	PCC(%)
2S1	**274**	0	0	0	0	0	0	0	0	0	100
BMP2	0	**195**	0	0	0	0	0	0	0	0	100
BRDM2	0	0	**271**	0	0	0	1	0	1	1	98.91
BTR60	0	0	0	**189**	1	0	0	0	0	5	96.92
BTR70	0	0	0	0	**196**	0	0	0	0	0	100
D7	0	0	0	0	0	**274**	0	0	0	0	100
T62	0	0	0	0	0	0	**271**	0	0	0	100
T72	0	0	0	0	0	0	0	**195**	0	0	100
ZIL131	0	0	0	0	0	0	0	0	**274**	0	100
ZSU23/4	0	0	0	0	0	0	0	0	0	**274**	100
Total											99.59

TABLE VI
PERFORMANCES OF DIFFERENT METHODS UNDER SOC

Input	Reference methods	Abbrev.	Feature	Fusion	Classifier	PCC (%)
Complex data	Bayesian Feature Matching [24]	BFM	Scattering center	No	Matching	95.62
	Sequential Feature Matching [61]	SFM	Scattering center	No	Matching	95.24
	Proposed ASC-BOVW	ASC-BOVW	Scattering center	No	RF	92.28
Amplitude image	Sparse Representation Classification [62]	SRC	RP features	No	SRC	96.29
	Support vector machine [9]	SVM	PCA features	No	SVM	96.78
	A-ConvNet [11]	ACN	Intensity values	No	softmax	99.13
	LM-BN-CNN [26]	LM-BN-CNN	Intensity values	No	softmax	96.44
	Proposed MVGGNet	MVGGNet	Intensity values	No	softmax	99.27
Complex data and amplitude image	Hierarchically Fusing Global and Local Features [24]	FGL	Scattering center and RP features	Decision fusion	Matching +SRC	99.08
	Tri-Task Joint Sparse Representation [63]	TJSR	Intensity, phase and Orientation	Feature fusion	SRC	98.05
	Proposed FEC	FEC	Scattering center and Intensity	Feature fusion	RF	99.59

CNN, the original target data are cropped to a size 128 × 128. The detailed recognition results of the proposed FEC model are shown in Table V. One can see that the PCCs of eight types of targets in all categories are 100% (except BRDM2 and BTR60). The overall PCC of FEC is 99.59%, which represents that most targets in the testing data are correctly classified.

To validate the recognition performance of the proposed FEC, a few published methods are chosen as comparison methods. The algorithm descriptions and corresponding recognition results of these methods are shown in Table VI. According to the input source, these methods are classified as three types: complex data-based methods, amplitude image-based methods, and the combined methods. As for the complex data-based methods, Bayesian feature matching (BFM) [24] and sequential feature matching (SFM) [61] work by matching the scattering centers in the pixel space. The PCCs of these two methods are 95.62% and 95.24%, respectively. To validate the effectiveness of the proposed scattering center feature extraction method based on the BOVW (ASC-BOVW), we also train an RF classifier to test the classification performance. As shown in Table VI, although the PCC of the proposed ASC-BOVW is lower than that of other scattering center-based methods, the ASC-BOVW can effectively transform the extracted scattering center set into a feature vector with little performance degeneration (approximately 3%). For the amplitude image-based methods, the RP-based SRC [62], the PCA-based SVM [9], A-ConvNet (ACN) [11], and large-margin softmax (LM-softmax) batch-normalization CNN (LM-BN-CNN) [26] are compared with the proposed MVGGNet. One can see that the PCC of the MVGGNet is 99.27%, which is higher than that of other amplitude image-based methods. In addition to the above single feature-based methods, different feature fusion-based methods are also compared in Table VI. The fusing global and local features (FGL) method [24] based on scattering centers and RP features achieve a PCC of 99.08%. Tritask joint SR (TJSR) [63] achieves a PCC of 98.05%. Compared with these methods, the feature fusion based on the scattering center and MVGGNet makes FEC the most powerful target recognition model with the highest PCC of 99.59%. Thus, the FEC successfully inherits the excellent performance of the scattering center feature and deep CNNs.

There are two critical steps in the proposed FEC framework: feature extraction and classification. To validate the recognition performance of different feature sets and classifiers, we have combined four features (PCA, ASC-BOVW, MVGGNet, and the FEC features) and four classifiers (SVM, KNN, SRC, and RF). The results of these methods are shown in Table VII. It can be seen that the dimension of the PCA feature is much larger than those of other types of features, increasing the overfitting risk of the classifier. The DCA method used in this article converts the two feature sets (ASC-BOVW and MVGGNet) into an 18-D feature vector, which is beneficial for training a robust classifier.

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TABLE VII
TRAINING AND TESTING IMAGES FOR THE SOC EXPERIMENTAL SETUP

Feature Type		PCA	ASC-BOVW	Modified VGGNet	Fused Feature
Feature Dimension		1439	256	256	18
Classifier	SVM	93.08	82.45	98.43	97.28
	KNN	89.63	86.64	98.85	97.48
	SRC	86.20	85.38	98.07	96.21
	RF	90.27	92.28	98.98	99.59

TABLE VIII
RECOGNITION RESULTS OF THE PROPOSED FEC UNDER DIFFERENT DEPRESSIONS

Dep.	Class	Results			PCC (%)	Average (%)
		2S1	BRDM2	ZSU23/4		
15°	2S1	273	0	1	99.64	99.52
	BRDM2	0	274	0	100	
	ZSU23/4	2	1	271	98.91	
30°	2S1	283	2	3	98.61	99.19
	BRDM2	2	285	0	99.30	
	ZSU23/4	0	1	287	99.65	
45°	2S1	238	21	44	78.55	81.08
	BRDM2	27	253	23	83.50	
	ZSU23/4	38	19	246	81.19	

For classifiers, the parameter setting of the SVM and SRC are the same as with [18]. K is set as 20 in the KNN method. The depth and number of decision trees in the RF are 12 and 300, respectively. The MVGGNet feature usually performs better than other types of features. It is noticeable that the combination of FEC fused features and the RF has the highest PCC compared to other methods. The features fused by FEC can be effectively discriminated in a low-dimension feature space, which makes these features easier to be combined with the RF classifier [60].

D. Results Under EOCs

There is a difference between the target data under different EOCs, as shown in Fig. 8. To validate the robustness of the proposed FEC, the EOC samples in Section III-A are used to train and test different recognition methods. BFM [24], SRC [62], SVM [9], ACN [11], TJSR [63], and FGL [24] are used as the comparison methods. The hyperparameters in these methods are consistent with those in the original articles [24]. When training the FEC under EOCs, the hyperparameters, e.g., the number of clustering centers, are the same as those under SOCs. The recognition results of these methods are given as follows.

1) Results Under EOC-1 (Large Depression Variant): The distributions of the training samples and testing samples are shown in Table III. Different from the samples under SOCs, there are three types of targets (2S1, BMP2 and ZSU23/4) under EOC-1. The training samples are acquired at a 17° depression. The testing samples are acquired at 15°, 30°, and 45°. The confusion matrices of the proposed FEC are shown in Table VIII. One can see that most targets under 15° and 30° are classified correctly. However, for the targets under 45°, the PCC of the proposed FEC decreases to 81.08% because the feature of the target data at 45° is different from those at 15° and 30°, as shown in Fig. 8. The recognition results of different methods are shown in Table IX. It can be seen that the scattering center-based methods perform better with large depressions than other amplitude image-based methods. Both the FGL and FEC methods improve the recognition PCC. For the proposed FEC method, the PCC at 30° is 99.19% and the PCC at 45° is 81.08%, which prove the effectiveness of the FEC method when the depression varies.

TABLE IX
RECOGNITION RESULTS OF DIFFERENT METHODS UNDER EOC-1

Method		PCC (%)		
		15°	30°	45°
Scattering Center	BFM	98.42	97.75	80.14
	ASC-BOVW	95.83	93.65	73.52
Amplitude Image	SRC	98.42	94.79	56.66
	SVM	98.90	98.42	67.00
	ACN	99.12	97.42	64.17
	MVGGNet	99.38	98.21	71.16
Fusion	TJSR	98.04	95.24	70.73
	FGL	99.14	99.07	83.39
	FEC	99.52	99.19	81.08

TABLE X
RECOGNITION RESULTS OF THE PROPOSED FEC UNDER EOC-2

Class	Serial Number	Results				PCC (%)
		BMP2	BRDM2	BTR70	T72	
BMP2	9566	425	2	0	1	99.30
	c21	421	3	3	2	98.14
T72	812	4	2	1	419	98.36
	A04	1	3	0	569	99.30
	A05	7	4	2	560	97.73
	A07	5	2	3	563	98.25
	A10	3	6	0	558	98.41
Average PCC (%)		98.48				

TABLE XI
RECOGNITION RESULTS OF DIFFERENT METHODS UNDER EOC-2 (%)

Method	BFM	SVM	ACN	SRC	FGL	FEC
PCC (%)	97.90	96.64	98.12	95.12	98.46	98.48

2) Results Under EOC-2 (Version Variant): The data in Table IV are used for training and testing the proposed methods. It is shown that BMP2, BRDM2, BTR60, and T72 are used as the training data and only BMP2 and T72 are used as the testing data. The recognition results of the proposed FEC method are shown in Table X. Although a few samples in some versions are wrongly classified, most of the targets in Table X are correctly classified. The recognition results of different methods when the version varies are shown in Table XI. All these methods can achieve PCCs beyond 95%. The PCC of the FGL method is higher than those of other traditional methods based on a single feature. The proposed FEC method achieves the highest PCC at approximately 98.48% by fusing the ASC feature and CNN feature. Therefore, the fusion strategy in the proposed FEC still can improve the recognition performance when the version varies.

3) Results Under EOC-3 (Noise Corruption): To evaluate the recognition performance of the proposed FEC method under noise corruption, the SOC data in the MSTAR data are used as the training data. Different levels of noise are added to the SOC testing data, as shown in Section III-A.

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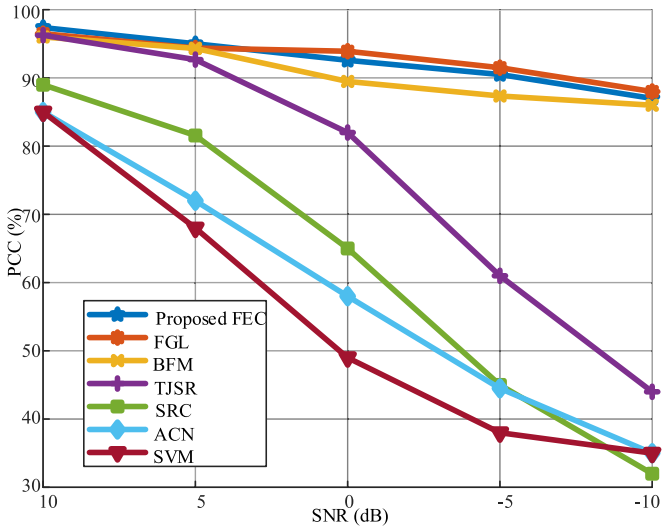


Fig. 10. Performances of different methods at different noise levels.

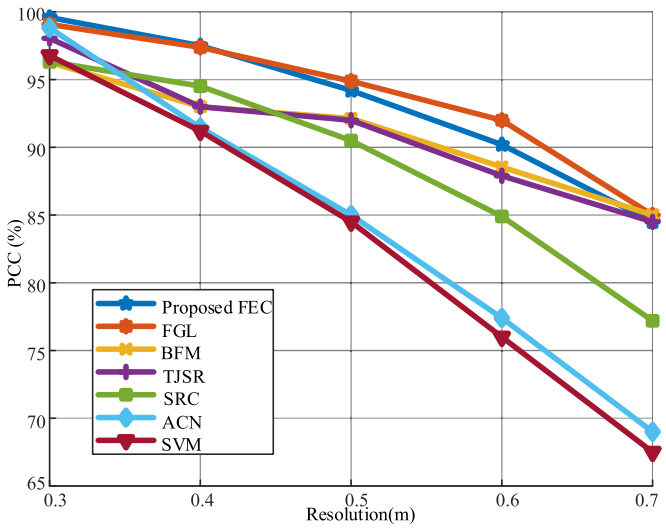


Fig. 11. Performances of different methods at different resolutions.

The recognition PCCs of the different methods with ten classes of targets are shown in Fig. 10. The BFM methods achieve better PCCs than other traditional methods at any noise level. This is because the scattering center can be effectively extracted from the noise-added data. After adding more noise, the PCC of the ACN method severely decreases because it is sensitive to the feature variants caused by noise. By fusing the ASC feature and SRC feature, FGL achieves the highest PCC among these methods. It is noticeable that the proposed FEC method can also achieve a high PCC at different noise levels. We believe that the fused feature in FEC does not only achieve good stability such as the ASC feature but it can also actively select the effective information of the CNN feature and utilize it.

4) Results Under EOC-4 (Resolution Variant): Like the experimental results under noise corruption, the SOC data in MSTAR are used as the training samples. The SOC testing data with resolution conversion in Section III-A are used as the testing data under varying resolutions. The recognition PCCs of different methods are shown in Fig. 11. The PCCs of all methods decrease seriously when the resolution decreases. The SRC, can, and SVM methods achieve low PCCs since these methods are sensitive to the image feature of amplitude images. The FGL method achieves the highest PCC at all the resolutions since it fuses the global feature and local feature. The proposed FEC method can also achieve a better recognition result than other recognition methods based on a single feature because it can effectively fuse the ASC feature and CNN feature using the DCA method.

TABLE XII
TRAINING AND TESTING IMAGES FOR OUTLIER REJECTION

Class	Serial Number	Training		Testing	
		Dep.	Number	Dep.	Number
BMP2	9563	17°	233	15°	195
BTR70	c71	17°	233	15°	196
T72	132	17°	232	15°	196
ZIL131	E12	-	-	15°	274
2S1	B01	-	-	15°	274

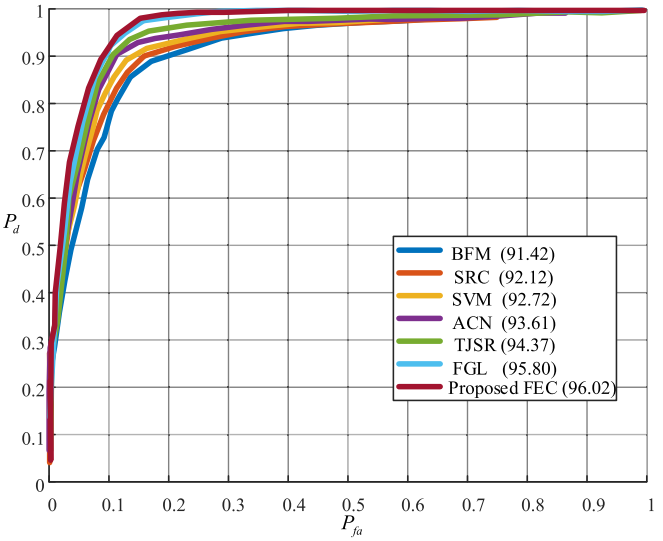


Fig. 12. ROC curves of different methods.

E. Results on Outlier Rejection

The above experiments on SOC data and EOC data perform "close-set" classifications. In this type of classification, the category of the testing data corresponds to that of the training data. However, when they are applied, the category of the testing data may not be included in the training data, which is named "open-set" classification. To evaluate this kind of classification in different methods, the training data and testing data are separated from the SOC data in the MSTAR data set, as shown in Table XII. The experimental setups of the different methods are consistent with those in Section III-B. A test sample is either identified to be a target or a confuser according to the threshold τ_{th} of the posterior probabilities. We define the detection rate P_{d} as the ratio of the number of known targets detected to the number of known targets in the testing set, and the false alarm rate P_{fa} as the ratio of the number of confusers declared as known to the number of confusers in the testing set. Then, the receiver operating

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characteristic (ROC) curves of the different methods are shown in Fig. 12. One can see that the proposed FEC method has a higher detection rate than other methods when a constant false alarm rate, e.g., $P_{\text{fa}} = 0.5$, is used. The FEC also has a lower false alarm rate than other methods at a constant detection ratio. Meanwhile, FEC achieves the best performance in outlier rejection with the maximum average precision of 96.02%. Therefore, for the testing samples that are not included in the training set, the proposed FEC can effectively discriminate their categories.

IV. Conclusion

The active recognition of SAR targets has been getting refined with the development of the electromagnetic scattering theory, machine learning, and image processing methods. Although many methods achieve highly accurate recognition for images under SOCs, the accurate recognition of SAR images under EOCs is still a challenge. Inspired by the good performance of the CNN and the excellent robustness of the scattering center, a feature fusion framework based on the CNN feature and scattering center feature is proposed in this article and is named the FEC. For the scattering center feature, we first extract the scattering centers from the input SAR complex data, then construct a BOVW from these scattering centers, and finally transform the extracted feature sets into feature vectors. Immediately afterward, these vectors can be used for feature fusion and classification. For the CNN feature, we propose an MVGGNet, where the number of fully connected layers is reduced according to the ASC feature. This MVGGNet does not only extract powerful features from the input amplitude image but it also achieves state-of-the-art recognition accuracy. For the feature fusion, the DCA method is introduced to the FEC framework. First, the correlation of the corresponding features across CNN features and ASC features is maximized; and second, the features belonging to different categories within each feature set are decorrelated. The experimental results on the MSTAR data set prove that the proposed FEC method attains higher recognition accuracy than any other methods under SOCs. Furthermore, FEC also performs well for SAR target data under different EOCs, e.g., large depressions and noise corruption. Therefore, the FEC method has good effectiveness and robustness, and great potential in the application of an SAR recognition system.

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